

MI-25: Partial Result

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Repository status: Partially resolved. The complete argument received an independent Codex-agent PASS review for the stated partial result only. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. [Detailed review and scope](#).

1 MI-25, endpoint only: the trace-norm Hlawka constant is infinite

For a norm $\|\cdot\|$, define the three-summand defect and the sum of pairwise defects by

$$\begin{aligned} N(X_1, X_2, X_3) &= \sum_{j=1}^3 \|X_j\| - \|X_1 + X_2 + X_3\|, \\ D(X_1, X_2, X_3) &= \sum_{1 \leq i < j \leq 3} (\|X_i\| + \|X_j\| - \|X_i + X_j\|). \end{aligned}$$

The full repository problem asks for the optimal dimension-independent constants C_p in $N \leq C_p D$ for Schatten norms. The result below resolves *only the endpoint* $p = 1$, which was left open in [1, Problem 7]. It does not determine C_p for $1 < p < \infty$.

Theorem 1.1. *No finite constant C_1 controls the three-summand trace-norm defect by the sum of pairwise trace-norm defects. This failure already occurs for real 2×3 rank-one matrices, or for real square 3×3 matrices after zero padding.*

Proof. Let

$$u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad u_2 = \begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix}, \quad u_3 = \begin{pmatrix} -1/2 \\ -\sqrt{3}/2 \end{pmatrix}.$$

These unit vectors satisfy

$$\sum_j u_j = 0, \quad \sum_j u_j u_j^T = \frac{3}{2} I_2, \quad u_i^T u_j = -\frac{1}{2} \quad (i \neq j).$$

For $0 < t < 1$, put

$$v_j = \begin{pmatrix} \sqrt{1-t} u_j \\ \sqrt{t} \end{pmatrix} \in \mathbb{R}^3, \quad X_j = u_j v_j^T \in \mathbb{M}_{2,3}(\mathbb{R}).$$

Each X_j has its only nonzero singular value equal to one. Furthermore,

$$\sum_j X_j = \begin{pmatrix} \frac{3}{2} \sqrt{1-t} I_2 & 0 \end{pmatrix}, \quad \left\| \sum_j X_j \right\|_1 = 3\sqrt{1-t}.$$

For $i \neq j$, write $U = (u_i \ u_j)$ and $V = (v_i \ v_j)$. The two Gram matrices are

$$U^T U = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}, \quad V^T V = \begin{pmatrix} 1 & (3t-1)/2 \\ (3t-1)/2 & 1 \end{pmatrix}.$$

They commute and are simultaneously diagonalized by $(1, 1)^T$ and $(1, -1)^T$. The squared singular values of $UV^T = X_i + X_j$ are therefore

$$\frac{1+3t}{4}, \quad \frac{9(1-t)}{4}.$$

It follows that every pair sum has trace norm

$$q(t) = \frac{1}{2}\sqrt{1+3t} + \frac{3}{2}\sqrt{1-t}.$$

Consequently,

$$\frac{N(X_1, X_2, X_3)}{D(X_1, X_2, X_3)} = \frac{1 - \sqrt{1-t}}{2 - q(t)}.$$

The denominator is positive for $t > 0$: strict concavity of the square root gives $\frac{1}{4}\sqrt{1+3t} + \frac{3}{4}\sqrt{1-t} < 1$. Expansion at zero gives

$$1 - \sqrt{1-t} = \frac{t}{2} + O(t^2), \quad 2 - q(t) = \frac{3}{4}t^2 + O(t^3).$$

Thus the defect ratio is asymptotic to $2/(3t)$ and tends to $+\infty$.

Alternatively, the divergence can be checked without Taylor expansions. If $g(t) = \sqrt{(1-t)(1+3t)}$, rationalization gives

$$2 - q(t) = \frac{6t^2}{(1+t+g(t))(2+q(t))}$$

and hence

$$\frac{1 - \sqrt{1-t}}{2 - q(t)} = \frac{(1+t+g(t))(2+q(t))}{6t(1+\sqrt{1-t})} \sim \frac{2}{3t}.$$

Appending a zero row leaves every singular value and every defect unchanged, giving the square-matrix assertion. $\square$

References

- [1] Koenraad M. R. Audenaert and Fuad Kittaneh. Problems and conjectures in matrix and operator inequalities. arXiv:1201.5232, 2012. Version consulted: v3; Problem 5, printed page 3, and Problem 7, printed pages 12–13. URL: <https://arxiv.org/abs/1201.5232>.